



Enhanced YOLO-Based Mango Detection and Centroid Tracking for Precise Yield Estimation

Hari Chandana Pichhika¹ · Priyambada Subudhi² · Raja Vara Prasad Yerra¹

Received: 7 February 2025 / Accepted: 27 April 2025 / Published online: 6 June 2025

© The author(s), exclusively licensed to Springer-Verlag GmbH Germany, a part of Springer Nature 2025

Abstract

Accurate mango yield estimation is crucial for efficient resource management and harvest planning. This study presents a novel framework utilizing a modified version of MangoYOLO5, a custom YOLOv5 model for mango detection proposed in our previous work. It includes an attentive feature fusion mechanism in the backbone, a convolutional block attention module, dense connections in the neck, and an additional 10×10 detection layer to enhance detection precision and reduce false positives. Furthermore, we integrate the proposed detection model with an enhanced centroid tracking algorithm that employs multi-frame references for occlusion handling and dynamic threshold adjustments to achieve consistent tracking and accurate fruit counting. The system's effectiveness is validated across varying light conditions and different trees, demonstrating close alignment with ground truth data from manual labeling, achieving a mean average precision (mAP) of 93.6%. This work not only advances the state-of-the-art in mango detection and yield estimation but also provides insights into developing robust computer vision solutions for agricultural applications.

Keywords Attention mechanism · Deep learning · Fruit counting · Precision agriculture · Object tracking

Introduction

Mangoes are highly valued in many tropical and subtropical areas worldwide due to their popularity and economic importance (Delhi 2018). Efficient mango farming requires meticulous planning and management including the early and precise prediction of yield, which is helpful in proactive planning for storage, packing, and other logistical considerations. Traditional manual counting methods are both labor-intensive and time-consuming (Anderson et al. 2019), with inherent inaccuracies arising from the repetitive counting.

To address these issues, the agricultural sector is increasingly adopting computer vision techniques for automated mango yield estimation. This involves collecting data in the form of images or videos, upon which deep learning (DL) models are trained. These models first detect and then count the detected mangoes. The accuracy of mango detection can be influenced by several factors, including changing lighting conditions and obstructions from elements such as fruits, leaves, or stems on the tree. In such cases, detection can be precise; the counting process may encounter issues, like counting the same mango multiple times, especially when collecting video data from various angles around the tree (Stein et al. 2016; Wang et al. 2019). Initially, for determining the yield from image or video datasets collected from mango farms, traditional feature-based techniques were employed. These methods involve extracting features such as color, shape, and texture using approaches like edge detection, blob analysis, and the randomized Hough transform (Neethi et al. 2023; Payne et al. 2013). However, these methods faced limitations, such as sensitivity to variations in lighting, issues with occlusion and overlap. As a result, the shift towards machine learning (ML) methods, including support vector machines (SVM), multi-layer perceptron, and random forests (Cengel et al. 2024; Neethi et al. 2021), offered improvements in detec-

Hari Chandana Pichhika
harichandana.p@iiits.in

✉ Priyambada Subudhi
priyambada.s@iiits.in

Raja Vara Prasad Yerra
yrv.prasad@iiits.in

¹ Electronics and Communication Engineering, Indian Institute of Information Technology, 517646 Sri City, Chittoor, Andhra Pradesh, India

² Computer Science and Engineering, Indian Institute of Information Technology, 517646 Sri City, Chittoor, Andhra Pradesh, India

tion and yield estimation accuracy. Despite their potential, ML models faced challenges when confronted with substantial variations in input data, such as changes in mango color, occlusion, and camera angles.

The introduction of DL techniques (Ergün 2024) has marked significant improvements in the field of agriculture. These methods automatically extract features from data, overcoming the limitations of manual feature design and showing superior performance under diverse conditions. While DL algorithms have shown promising results, a significant limitation is that most of the previously proposed methods have considered one side of the tree for yield estimation (Neethi et al. 2023). However, this approach may not provide an accurate yield estimate, as mangoes can be distributed all around the tree. Recent studies have aimed to improve accuracy by considering multiple sides, employing either DL-based tracking algorithms (Li et al. 2023; Li et al. 2024; Wang et al. 2019), 3D-based reconstruction methods (Stein et al. 2016), and transformer-based approaches (Lin et al. 2021; Zhang and Tian 2023). Although tracking algorithms like SORT (Bewley et al. 2016) and Deep SORT (Wojke et al. 2017) have proven effective for various fruits like apples, citrus, and pineapples, applying these methods to static objects of similar shape and size, such as mangoes, presents unique challenges. These challenges primarily arise from the limited variability in object features, which reduces the effectiveness of the algorithms in such scenarios. Moonrinta et al. (2010) developed an automated system for mapping and predicting pineapple crops using video cameras, employing Speeded-Up Robust Features (SURF) feature points for precise fruit detection. Similarly, Chaivivatrakul et al. (2010) introduced a novel approach for pineapple detection and 3D reconstruction from video sequences, utilizing Harris corner detection, Scale-Invariant Feature Transform (SIFT) keypoint descriptors, and SVM for keypoint classification, achieving significant accuracy in fruit dimension estimation.

In another study by Santos et al. (2020) focused on grape yield prediction and automated harvesting using convolutional neural networks (CNNs), achieving a high F1-score of 0.91. Also, the authors introduces a public dataset with annotated grape clusters and a novel annotation methodology for segmenting complex objects in natural images. Parico and Ahamed (2021) introduce a real-time pear fruit counter for mobile applications using You Only Look Once version 4 (YOLOv4) and Deep SORT tracking, achieving an F1 count of 87.85%, attributed to YOLOv4s low false negatives. Later on, Junos et al. (2022) proposed an improved You Only Look Once (YOLO) model for detecting oil palm loose fruits from unmanned aerial vehicle (UAV) images. Key improvements include a densely connected network for feature reuse, swish activation, multi-layer detection for small targets, and prior box optimization for

accurate bounding boxes. Similarly, Prakash and Prakasam (2023) introduced a CNN-based intelligent fruit classifier using bilinear pooling with heterogeneous streams, treating fruit classification as a fine-grained visual classification (FGVC) problem. Zhang et al. (2022) enhanced the YOLOv3 network into OrangeYOLO for detecting small citrus fruits at various scales, incorporating a channel and spatial attention multi-scale fusion module. To address double-counting of occluded fruits, they proposed the OrangeSort algorithm, which uses a tracking region counting strategy and a tracking algorithm based on motion displacement estimation. Wang et al. (2022) introduced a litchi fruit detection method with an improved YOLOv5s model, achieving a 62.77% mean average precision. In Li et al. (2024), the authors developed a lightweight apple detection model by modifying YOLOv5s with a 3×3 Conv and ShuffleNetV2 backbone, removing the SPP and C3 layers, using DWConv in the neck, and replacing PANet's Concat layers with Add layers, resulting in a more efficient and compact model. Similarly, Si et al. (2024) proposed a novel registration algorithm for aligning thermal infrared and visible apple images by integrating feature point registration with an active contour model, improving the accuracy of multi-modal image alignment for agricultural applications. Zhao et al. (2024) proposed YOLO-Granada, an enhanced version of YOLOv5 designed for real-time pomegranate detection. This model integrates ShuffleNetv2, grouped convolution, channel shuffle, and the Convolutional Block Attention Module (CBAM), achieving 92.2% accuracy and a 17.3% improvement in detection speed. We have incorporated this technique in our comparative analysis to evaluate its performance relative to other methods.

We have implemented an improved, fast, and compact version of YOLOv5s (termed MangoYOLO 5) (Hari Chandana et al. 2023) for efficient detection in MangoNet Semantic Dataset (Kestur et al. 2019). Despite its effectiveness, it exhibits few limitations, including false positives and inaccuracies in bounding box localization for partially occluded mangoes. Among the approaches for yield estimation, two primary methods are 3D reconstruction (Stein et al. 2016) and tracking algorithms (Zhang et al. 2022). The 3D reconstruction method captures detailed spatial distribution and provides a comprehensive tree representation but requires specialized equipment such as multi-camera setups or Light Detection and Ranging (LiDAR) sensors, and involves complex data processing, increasing both cost and computational overhead, making it less practical for certain conditions. In contrast, tracking algorithms are cost-effective, computationally efficient, and rely on simple video-based inputs. While they may face challenges with overlapping objects, they are well-suited for this study, which involves static objects and simple configurations, enabling effective tracking and yield estimation. Based on these con-

siderations, the tracking algorithm was selected as it aligns better with the requirements of this study. This study builds on these insights by proposing architectural enhancements to improve mango fruit detection accuracy and yield estimation, particularly addressing limitations in complex field conditions. In this work, we proposed a novel DL framework focusing on mango detection and yield estimation. The key contributions of this work are summarized below:

- i) We provide a carefully tailored mango video dataset, offering a complete 360° coverage of each tree.
- ii) We proposed an enhanced MangoYOLO5 model featuring attentive feature fusion in the backbone, CBAM-enhanced dense connections in the neck, and an additional 10×10 detection layer for enhanced accuracy in mango detection.
- iii) A multi-frame reference based centroid-tracking algorithm is also proposed with dynamic threshold selection mechanism to improve generalizability of the approach.

The structure of the remaining sections in this article is outlined below. Section 2 details the proposed framework. Experimental results are presented in Sect. 3, followed by a discussion based on our findings. Finally, conclusions and future directions are presented in Sect. 4.

Methodology

The methodology followed in the proposed work is discussed precisely in this section, and the various steps involved here are depicted in Fig. 1. The following subsections briefly discuss each of these steps in detail.

Data Collection and Pre-processing

A comprehensive video dataset was created by recording eight *Mangifera Indica* ('Banganapalle') mango trees in a 360° view using a handheld mobile camera with a resolution of 1920×1080 pixels. The recordings were conducted during the daytime in the tropical summer months of April–June 2022 at a mango orchard located at the IIIT Sri City campus (latitude: 13.54951°N, longitude: 80.02749°E), Andhra Pradesh, India. The camera was chosen for its accessibility and cost-effectiveness, with imaging performed at a consistent distance of 1.5–2 m to balance the field of view and resolution. The featured trees are medium-sized 'Banganapalle' mango trees with an average height of 2–3 m and a canopy diameter of 1–2 m, containing approximately 50–60 mangoes per tree visible under natural conditions. The orchard-style planting configuration, with a spacing of approximately 2 m between trees, facilitated optimal light penetration and smooth navigation for video recording. Each tree was recorded by walking around it in a 360° path, capturing an average 60–80 s of video per tree to ensure comprehensive canopy coverage under varying lighting conditions (sunny and partially cloudy). To enhance diversity, the dataset initially consisted of 704 image patches of size 256×256 pixels, extracted from three videos. Augmentation techniques, including rotation (90°, 180°, 270°), flipping (horizontal and vertical), and scaling, were applied to each image, generating six additional images per original. This resulted in a total of 5000 image patches for training. Additionally, around 400 high-resolution full-size images (1920×1080) were included for training and validation. Testing was performed on five separate videos. Since the original dataset lacked YOLO-compatible labels, all images were manually annotated using the makesense.ai tool to generate ground

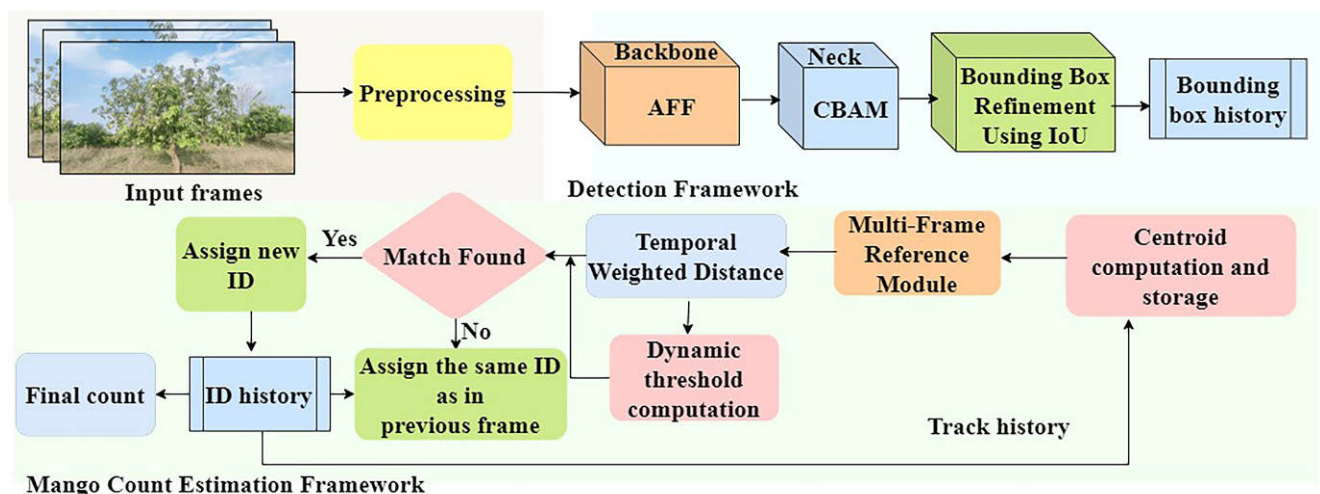


Fig. 1 Overview of the proposed mango detection and tracking framework. *AFF* Attentive Feature Fusion, *CBAM* Convolutional Block Attention Module, *IoU* Intersection over Union

truth bounding boxes. The dataset was divided into training and validation sets in an 80% and 20% ratio, while testing was conducted directly on videos.

Proposed Detection Model

The primary objective of the enhanced MangoYOLO5 model is to increase the precision and accuracy of detection, reducing false positives and addressing the key challenges in yield estimation tasks. While the baseline MangoYOLO5 model (Hari Chandana et al. 2023) significantly improves on the general YOLOv5s for mango detection, certain limitations such as false positives and inaccuracies in bounding box localization for occluded mangoes were observed. To tackle these issues, we propose a series of architectural enhancements, which are detailed in the following subsections.

Attentive Feature Fusion in the Backbone

We integrated a unique Attentive Feature Fusion (AFF) block (see Fig. 2) in the back-bone layer in our proposed architecture, designed to selectively emphasize important features during the feature extraction process in the backbone. The AFF block (Dai et al. 2021) consists of multiple attention mechanisms that work synergistically to boost the model's focus on relevant regions, ensuring that key features are effectively captured and utilized for detection.

The core components of the AFF block include:

- **Channel Attention (CA):** This submodule applies global average pooling and channel-wise dependencies to recalibrate features along the channel dimension, enhancing feature representations based on their significance.

- **Spatial Attention (SA):** This component generates spatial attention maps by leveraging both average pooling and max pooling along the channel axis, which are then combined to highlight important spatial regions in the feature map.
- **Cross-layer Feature Fusion (C2F):** To enrich feature information, C2F combines features from different layers, promoting the propagation of useful information across layers, which is crucial for capturing details in small object detection.
- **Multi-scale Feature Aggregation (MSFA):** The MSFA mechanism aggregates features at different scales to capture both global and local context, improving detection performance for objects of varying sizes.
- **Identity:** This acts as a direct skip connection, allowing the unaltered input features to pass through and be combined with the outputs of the other attention mechanisms, ensuring the retention of original feature information for better representation.

Each of these modules within the AFF block plays a role in selectively enhancing feature maps, allowing the model to focus on areas with occluded or clustered mangoes. The output of the AFF block is subsequently passed through convolutional layers, resulting in a more refined representation that enhances the backbone's ability to capture critical features.

CBAM and Dense Connections in the Neck

In the next section, we introduce the CBAM (Woo et al. 2018) and dense connections to further enhance feature flow and localization precision. CBAM integrates both channel and spatial attention mechanisms, enabling the network to adaptively emphasize informative regions in the feature

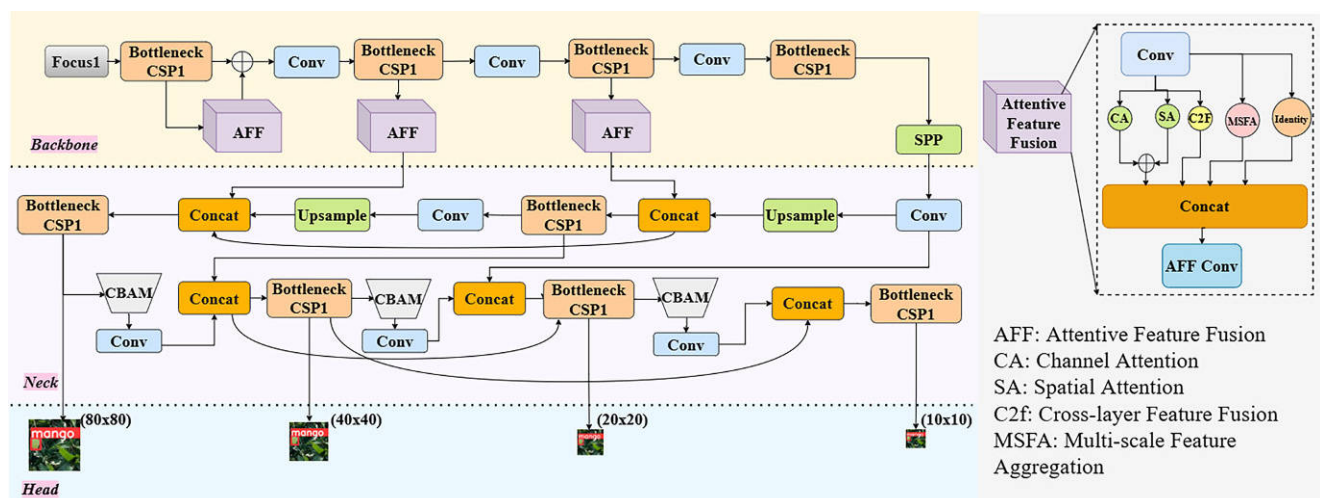


Fig. 2 Enhanced MangoYOLO5 architecture. *SPP* Spatial Pyramid Pooling, *CSP1* Cross Stage Partial Layer 1, *Conv* Convolution, *Concat* Concatenation

map. Its inclusion in the neck is essential because, after feature extraction in the backbone, the neck aggregates multi-scale features that may still contain irrelevant or redundant information. By applying CBAM again, the network adaptively refines these aggregated features, emphasizing important spatial and channel-specific details to ensure more accurate localization and context-aware detection at different scales. Meanwhile, the dense connections ensure efficient gradient propagation and feature reuse across layers, contributing to stronger feature representation without increasing the model's complexity excessively.

Enhanced Detection Head

To enhance mango detection and address limitations in the baseline model, an additional 10×10 detection head has been incorporated into the model's head section. This detection layer captures coarser features, making it particularly effective for detecting larger mangoes. It is integrated into a multiscale detection pipeline by fusing it with higher-resolution layers (20×20 , 40×40 , and 80×80). This hierarchical fusion enables the model to utilize contextual information at multiple scales, improving detection accuracy for mangoes of various sizes. The enhanced architecture strengthens the semantic connections between shallow and deep features, as illustrated in Fig. 2.

To further enhance detection accuracy, an intersection over union (IoU)-based refinement step is employed for bounding boxes. By setting a lower IoU threshold ($\tau_{\text{IoU}} = 0.5$) compared to the conventional threshold 0.7, the refinement reduces false positives and ensures tighter bounding box localization.

Proposed Centroid Tracking Algorithm

In the mango tracking stage, we employed an enhanced centroid tracking algorithm to accurately estimate the yield by monitoring the movement of detected mangoes across successive video frames. This process assigns a unique identifier (ID) to each mango for consistent tracking and incorporates several improvements, including multi-frame reference consideration, and dynamic threshold computation to the traditional centroid tracking approach as explained in the following subsections.

Multi-frame Reference-Based Centroid Tracking

The proposed tracking algorithm operates on each frame considering bounding box results generated by the enhanced MangoYOLO5 model. In any t^{th} frame, let us assume there are n_1 detected mangoes and the bounding box details for all such mangoes are obtained from the

bounding box history. Then for each i^{th} mango detected in the t^{th} frame, the centroid is computed as:

$$C_t^i(x, y) = \left(\frac{x_{\min} + x_{\max}}{2}, \frac{y_{\min} + y_{\max}}{2} \right) \quad (1)$$

where i varies from 1 to n_1 .

To verify it is the same mango as appeared in the previous frame, the distance between its centroid and that of the centroid of the previous frame's mango needs to be computed. However, to handle scenarios such as occlusions, where a mango may be hidden in one frame but visible in the previous frame, the algorithm considers references from both the immediate previous frame ($t-1$) and the second previous frame ($t-2$). For each i^{th} centroid in the current frame and j^{th} centroid in a previous frame, the temporal weighted distance is calculated as:

$$d_{ij}(C_t^i, C_{t-k}^j) = \frac{1}{k} \sqrt{(x_t - x_{t-k})^2 + (y_t - y_{t-k})^2} \quad (2)$$

where i varies from 1 to n_1 , j varies from 1 to n_2 (assuming there are n_2 mangoes detected in the previous frame), and k represents the temporal offset ($k=1$ for ($t-1$), $k=2$ for ($t-2$)th frame). The temporal weighting $\frac{1}{k}$ prioritizes centroids from recent frames, reducing the impact of occlusion. Although k can be extended to include more frames for greater accuracy, this increases computational complexity. To balance accuracy and efficiency, we have chosen $k=1, 2$ for this work.

A match occurs if the following condition is satisfied:

$$d_{ij} < \theta \quad (3)$$

where θ is dynamically updated based on recent centroid movements for each mango, as explained in the following subsection. It uses a historical buffer D_{hist} which stores the recent centroid distances, and is refreshed at the end of each frame to ensure only the most recent movements are considered. For every i^{th} mango in the current frame, distances are compared against all j^{th} mangoes in the ($t-1$)th frame and stored in the historical buffer. If the condition in Eq. 3 is satisfied for the i^{th} mango in the current frame, it is assigned the same ID as the j^{th} mango in the previous frame. If none of the j^{th} mangoes in the ($t-1$)th frame satisfy the condition, the algorithm computes centroid distances with the ($t-2$)th frame. If no match is found in either ($t-1$) and ($t-2$), the i^{th} mango in the current frame is assigned a new ID.

Once, a matching mango with a corresponding ID is identified, either from the immediate previous frame ($t-1$) or the second previous frame ($t-2$), then the algorithm further validates the match using bounding box overlap

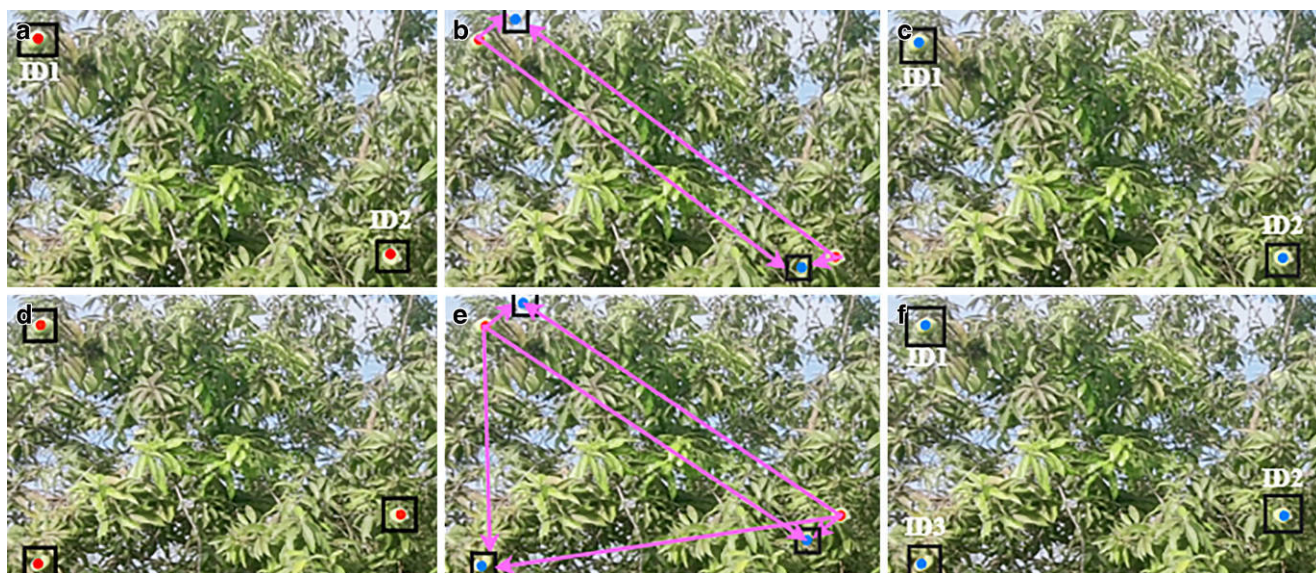


Fig. 3 Demonstration of working of the proposed centroid tracking algorithm across three consecutive frames. **(a)** Two mangoes detected (red dots) with IDs ID1 and ID2. **(b)** Temporal weighted distances computed; magenta lines indicate matching between centroids (blue dots) of detected mangi in the current frame and the previous frame. **(c)** IDs retained after matching. **(d)** New mango appears (red dot). **(e)** Matching across frames shown by magenta lines connecting centroids. **(f)** New mango assigned ID3 after failing threshold match

through IoU. For a detected bounding box $B_t^i = [x_b, y_b, w_b, h_b]$, the match condition is:

$$\text{Match}(B_t^i, B_{t-k}^j) = \begin{cases} \text{True, if } d_{ij} < \theta \text{ and } \text{IoU}(B_t^i, B_{t-k}^j) \geq \tau_{\text{IoU}} \\ \text{False, otherwise} \end{cases} \quad (4)$$

The working of the proposed tracking approach has been demonstrated in Fig. 3. Figure 3a shows a random frame with two mangoes detected and assigned IDs as ID1 and ID2. Figure 3b shows the next frame with two mangoes, where the temporal weighted distance is computed using the centroids of the detected mangoes from the previous frame. The IDs (ID1 and ID2) are retained in Fig. 3c after successful matching with their respective centroids. Figure 3d shows the third consecutive frame, where a newly visible mango appears. In Fig. 3e, the algorithm matches the visible mangoes in the current frame with those detected in the last two frames ($t-1$ and $t-2$), using temporal weighted distances and bounding box overlaps. Mangoes ID1 and ID2 maintain their original IDs. Finally, in Fig. 3f, the newly visible mango does not satisfy the threshold condition for matching based on previous frames. As a result, it is assigned a new ID (ID3), indicating a newly detected mango.

Dynamic Threshold Computation

To accommodate variations in mango positions and movements, the algorithm dynamically updates the threshold

θ used for centroid matching. The initial threshold is set to $\theta_{\text{init}} = 10$ pixels. For each i^{th} mango in the current frame, θ is calculated based on the mean and standard deviation of the centroid distances stored in the historical buffer D_{hist} , which retains recent displacement data:

$$\mu_t^i = \frac{1}{|D_{\text{hist}}|} \sum_{j=1}^{|D_{\text{hist}}|} d_j, \quad \sigma_t^i = \sqrt{\frac{1}{|D_{\text{hist}}|} \sum_{j=1}^{|D_{\text{hist}}|} (d_j - \mu_t^i)^2} \quad (5)$$

where d_j is the j^{th} centroid displacement in the buffer. The threshold is then updated dynamically using a smoothing factor ($\alpha = 0.7$) and a scaling multiplier ($k = 1.5$):

$$\theta = \alpha \cdot \theta_{t-1} + (1 - \alpha) \cdot (\mu_t + K \cdot \sigma_t). \quad (6)$$

To ensure stability, the threshold is bounded within a minimum ($\theta_{\text{min}} = 5$ pixels) and a maximum ($\theta_{\text{max}} = 20$ pixels):

$$\theta = \max(\theta_{\text{min}}, \min(\theta, \theta_{\text{max}})) \quad (7)$$

The proposed centroid tracking algorithm, detailed in Algorithm 1 (see Fig. 4), is applied iteratively over all frames in the video sequence to ensure robust mango tracking. As outlined in step 4, centroids for all detected bounding boxes in frame t are first computed. For each centroid (steps 5–17), the algorithm attempts to associate it with centroids from the two preceding frames ($t-1$ and $t-2$), as

Fig. 4 Algorithm 1: proposed centroid tracking algorithm

Require: Bounding boxes B_t for each frame t , initial threshold θ_{init} , IoU threshold τ_{IoU}

- 1: Initialize $\theta \leftarrow \theta_{\text{init}}$, $\theta_{\text{min}} \leftarrow 5$, $\theta_{\text{max}} \leftarrow 20$, $\alpha \leftarrow 0.7$, $\tau_{\text{IoU}} \leftarrow 0.5$
- 2: Initialize historical buffer D_{hist} and ID list
- 3: **for** each frame t **do**
- 4: Compute centroids $C_t[i]$ for each detected mango from $B_t[i]$
- 5: **for** each centroid $C_t[i]$ **do**
- 6: $\text{matched} \leftarrow \text{False}$
- 7: **for** $k = 1$ to 2 **do**
- 8: **for** each centroid $C_{t-k}[j]$ in frame $t-k$ **do**
- 9: Compute distance $d_{ij} \leftarrow \frac{1}{k} \sqrt{(x_t - x_{t-k})^2 + (y_t - y_{t-k})^2}$
- 10: **if** $d_{ij} < \theta$ **and** $\text{IoU}(B_t[i], B_{t-k}[j]) \geq \tau_{\text{IoU}}$ **then**
- 11: Assign $ID_t[i] \leftarrow ID_{t-k}[j]$
- 12: Update $D_{\text{hist}}[i]$ with d_{ij}
- 13: $\text{matched} \leftarrow \text{True}$
- 14: **break**
- 15: **end if**
- 16: **end for**
- 17: **if** matched **then**
- 18: **break**
- 19: **end if**
- 20: **end for**
- 21: **if not matched then**
- 22: Assign new ID to $C_t[i]$ and initialize $D_{\text{hist}}[i]$
- 23: **end if**
- 24: **end for**
- 25: **end for**

Ensure: Unique and consistent mango IDs across video frames

shown in the loop over $k=1$ to 2 (step 7), enabling a multi-frame reference mechanism. step 9 computes the temporal weighted Euclidean distance between centroids, while step 10 evaluates both this distance and the IoU overlap between corresponding bounding boxes against their respective thresholds (θ and τ_{IoU}). If a match is confirmed, the ID is retained (step 11), and the historical displacement buffer is updated (step 12). If no match is found across both reference frames, a new ID is assigned to the object (step 17). This structured approach reliably tracks mangoes across frames despite occlusion, false detections, or reappearances, thereby enhancing the reliability of mango count estimation.

Discussion of Results and Experimental Analysis

In this section, we present and examine the outcomes of our developed system for on-tree detection and tracking of mangoes by dividing the results obtained into two subsections: detection results and tracking results. Experiments were conducted on a high performance computing (HPC)

facility graphics node with Intel® Xeon® processor (Intel Corporation, Santa Clara, CA, USA) and an NVIDIA® Tesla® graphics processing unit (GPU) (NVIDIA Corporation, Santa Clara, CA, USA), featuring 16 GB memory, a base clock of approximately 1190 MHz, and 3584 CUDA cores. The evaluation of the algorithms, based on detection and tracking, is presented in detail in the subsequent subsections.

Detection Results

The proposed detection method demonstrates efficiency in handling various lighting conditions, which were naturally introduced as the dataset includes frames captured by circling the tree. As shown in Fig. 5, the proposed enhanced MangoYOLO5 model consistently detects all mangoes across different lighting scenarios. To further evaluate its performance, we compared the results of the proposed model with those of state-of-the-art models, including Faster R-CNN (Ren et al. 2015), YOLOv5s (Ultralytics, 2021), MangoNet (Kestur et al. 2019), the baseline MangoYOLO5 (Hari Chandana et al. 2023), and advanced YOLO series models such as YOLOv8, YOLOv9,

Fig. 5 Mango detection under different lighting conditions across random frames using enhanced MangoYOLO5 model

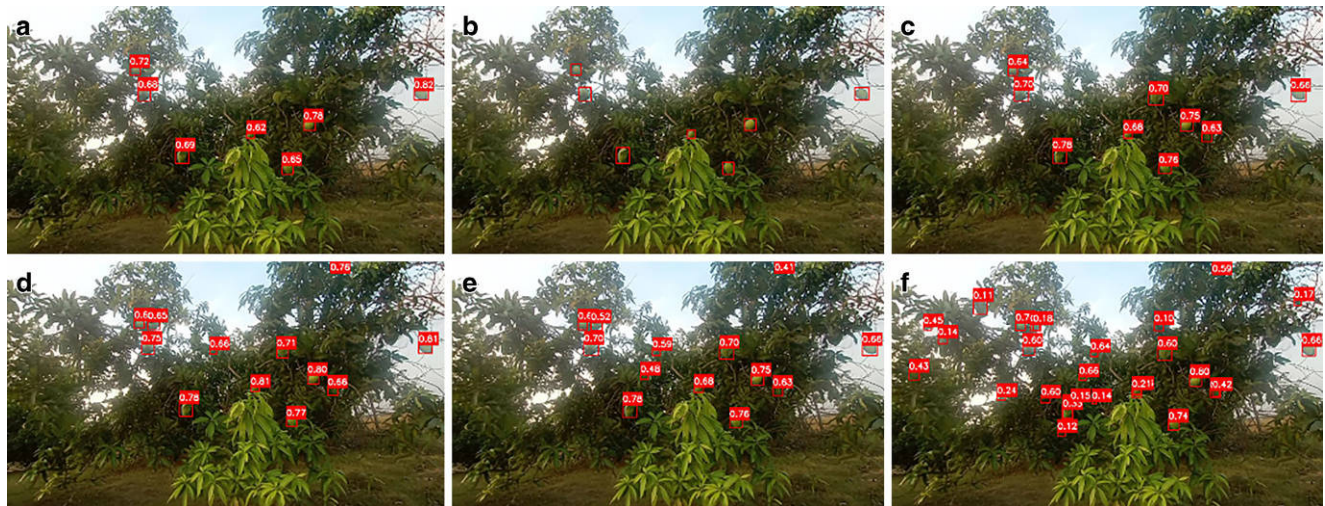
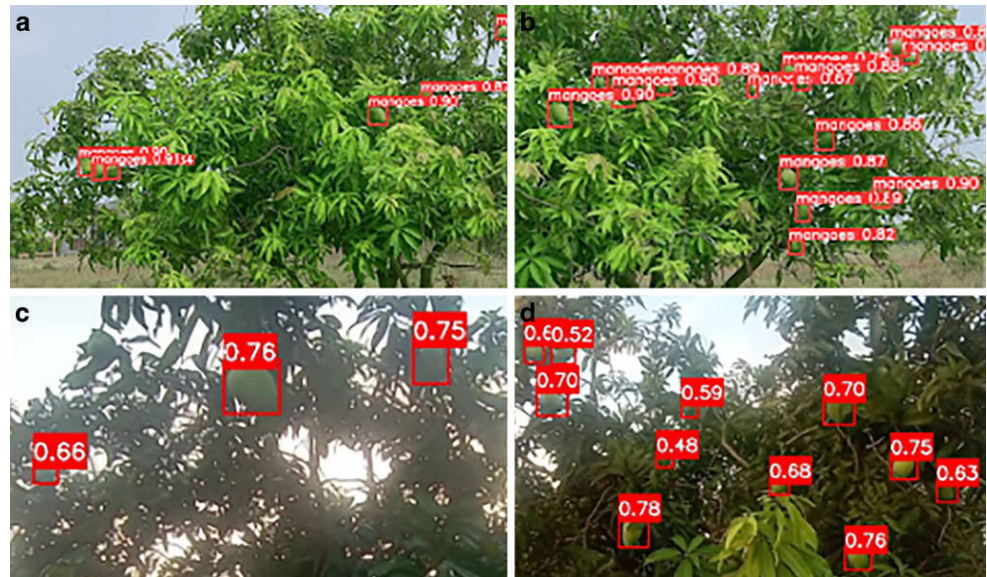


Fig. 6 Comparison of on-tree mango detection results on a random frame. **a** Faster R-CNN, **b** MangoNet, **c** YOLOv5s, **d** MangoYOLO5, **e** YOLOv11, **f** Enhanced MangoYOLO5

YOLOv10, and YOLOv11 (Sharma et al. 2024). All models were uniformly configured with 100 epochs, a learning rate of 0.01, the AdamW optimizer, and a batch size of 16. Under challenging lighting conditions, as illustrated in Fig. 6, the proposed model demonstrates superior detection capabilities, achieving higher probability scores even for obscured mangoes, outperforming the other models. MangoNet and YOLOv5s exhibit limitations in detecting several mangoes under these conditions, while YOLOv11, although effective, tends to produce more false positives. This highlights the robustness of the proposed model in handling occlusion and overlapping mangoes.

In our detailed analysis of detection outcomes, we conducted a quantitative comparison between the above-mentioned models using three key performance metrics:

precision, recall, and mean average precision (mAP). Precision, defined as the ratio of true positive detections to the total number of positive detections, and recall, the ratio of true positive detections to actual mangoes. The mAP provides an overall performance metric by averaging precision across multiple recall levels. The performance of the models is summarized in Table 1. The proposed enhanced MangoYOLO5 achieves the highest mAP (91.6%) compared to other models, demonstrating significant improvements in precision and recall over MangoYOLO5, YOLOv8, and YOLOv9. Although the estimation time (~30 min) is slightly higher than MangoYOLO5 (~22 min), the substantial gains in detection accuracy highlight the model's robustness in challenging conditions such as occlusions and scale variations. Additionally, the model

Table 1 Comparison of performance metrics for baselines and proposed model using patches (256×256 pixels)

Model name	Precision (%)	Recall (%)	mAP (%)	Estimation time (min)	FLOPs (G)	Parameters (M)
Faster R-CNN	74	48	59	~44	158.7	41.2
MangoNet	75.5	50	60.3	~160	250.4	1.11
YOLOv5s	85.5	58.9	68.5	~35	17.4	7.2
MangoYOLO5	79.5	68.4	81.2	~22	15.4	5.8
Mutual-Attention Transformer	86.8	72	88.5	~40	23.5	10.2
YOLOv8	87.5	75.2	89.8	~30	24.3	9.8
YOLOv9	88	76	90.2	~28	25.1	10.5
YOLO-Granada	88.2	74.5	90.4	~32	18.9	9
YOLOv10	89	76.5	91.4	~30	26	11.1
YOLOv11	89.5	77.2	91.7	~33	28.5	12.4
Enhanced MangoYOLO5	89.3	77	91.6	~30	19.6	8.5

mAP Mean average precision, *FLOPs* Floating-point operations per second

Table 2 Comparison of performance metrics for baselines and proposed model using full resolution images (1920×1080 pixels)

Model name	Precision (%)	Recall (%)	mAP (%)	Estimation time (min)	FLOPs (G)	Parameters (M)
Faster R-CNN	78	52	62.5	~55	205	41.2
MangoNet	78.8	55	63.8	~360	458.7	1.11
YOLOv5s	70.8	64.6	70.8	~75	22.5	7.2
MangoYOLO5	88.3	74.9	85.1	~42	26.4	5.8
Mutual-Attention Transformer	88	74.8	90	~55	33	10.2
YOLOv8	90.5	77	91.2	~45	27.8	9
YOLOv9	91	86.7	93.5	~42	30.5	9.8
YOLO-Granada	91.8	79.2	93	~40	32	10.5
YOLOv10	92.5	79.5	93.5	~44	33.5	11.1
YOLOv11	93	80.2	93.7	~47	36	12.4
Enhanced MangoYOLO5	91	79	92.8	~42	25.5	8.5

mAP Mean average precision, *FLOPs* Floating-point operations per second

balances computational efficiency and detection accuracy by leveraging 256×256 patches, which reduce resource requirements without compromising performance.

To evaluate the effect of using smaller patches, we also conducted experiments with full-sized images at a resolution of 1920×1080. The results, presented in Table 2, show that the enhanced MangoYOLO5 model achieved a slight improvement in mAP, increasing from 91.6% to 92.8% with full-resolution images. However, marginal gain came with a significant increase in computational demands, with floating-point operations per second (FLOPs) rising by approximately 30%, and estimation times increasing across all models. Patch-based processing proves beneficial for reducing computational complexity and parameter requirements, making it ideal for scenarios with resource constraints. However, in cases where computational resources are not a limiting factor, full resolution images can be utilized to achieve better accuracy.

Tracking Results

In the tracking algorithm, we applied an enhanced centroid-based tracking approach with a dynamic threshold to classify newly detected mangoes as either new or existing. The results of the proposed detection model were processed through widely adopted tracking algorithms such as SORT (Bewley et al. 2016), Deep SORT (Wojke et al. 2017), FairMOT (Y. Zhang et al. 2021), and the proposed tracking algorithm, as these methods are commonly used for yield estimation across various crops. We have observed that the SORT algorithm generates numerous false positives, counting the same mango multiple times and inflating the total count. Similarly, while Deep SORT utilizes a reference box for mango counting, challenges arise due to camera movement, leading to inaccuracies and an over-estimation of mangoes. In contrast, FairMOT (as shown in Fig. 7a) improves mango tracking by assigning unique IDs to mangoes, thereby reducing redundant counts. This improvement is evident in its ability to maintain consistent ID assignment across specific frames. However, our pro-

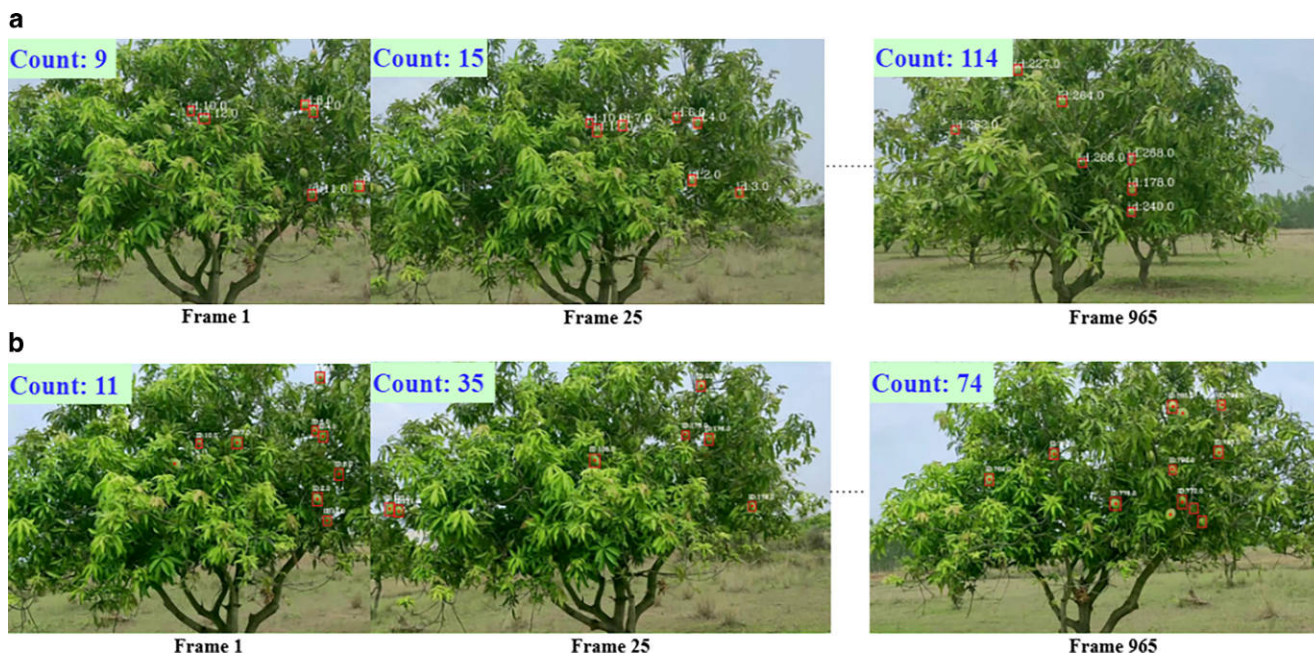


Fig. 7 Demonstration of detection cum tracking results for a single video using **a** Fair MOT, **b** proposed model

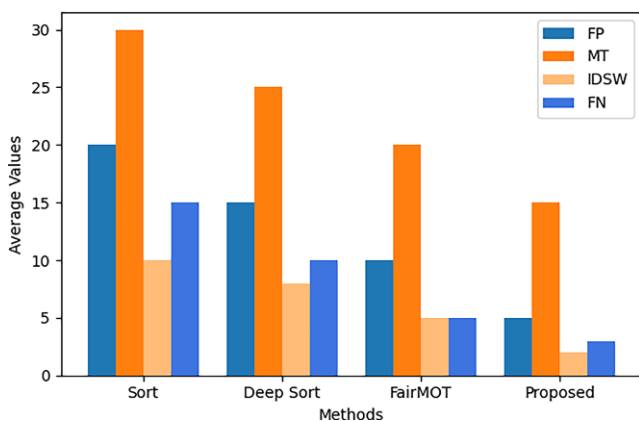


Fig. 8 Tracking performance in terms of correctly tracked instances, ID switches (IDSW), false positives (FP), and false negatives (FN) for SORT, Deep SORT, FairMOT, and the proposed model

posed algorithm (Fig. 7b) outperforms FairMOT, yielding significantly fewer redundant counts. It is important to note that IDs are assigned randomly during the tracking process, not sequentially. The final mango count is determined by counting unique IDs across frames and eliminating duplicates caused by reappearances, ensuring an accurate and reliable count.

For tracking the mangoes, we used the multi-object tracking accuracy (MOTA) score, defined in Eq. 8, which quantifies the accuracy of mango tracking by considering the sum of false negatives (FN), false positives (FP), and identity switches (IDSW), normalized by the total number of ground truth objects (GT). Figure 8 shows the average tracking performance metrics for SORT, Deep SORT,

FairMOT, and the proposed method across five videos, including FP, FN, IDSW, and Mostly Tracked (MT). FP represents non-mango objects incorrectly identified as mangoes, FN counts actual mangoes missed, IDSW indicates incorrect ID switches, and MT shows successfully tracked mangoes. The proposed method performs best overall, with the lowest FP and IDSW, and the highest MT, indicating robust tracking accuracy.

$$\text{MOTA} = 1 - \frac{\sum_t (\text{FN}_t + \text{FP}_t + \text{IDSW}_t)}{\sum_t \text{GT}_t} \quad (8)$$

To evaluate the overall yield estimation performance of the proposed approach, we utilized a ground truth count (GT), determined through manual counting of mangoes in the video frames during the labeling process. Table 3 displays the comparison results of the described methods for a series of five videos. It includes the GT counts along with the corresponding errors relative to the GT values. The proposed method reported lower mean absolute error (MAE) values, indicating improved counting accuracy in most cases. While FairMOT achieved mango counts numerically closer to the ground truth in videos 1 and 3, it produced overestimated counts in challenging visual conditions, particularly under inconsistent lighting and motion blur—as reflected by its lower MOTA values. In contrast, although the proposed approach shows a slightly higher MAE in some cases, it consistently achieved higher MOTA values (0.79 and 0.80 vs. FairMOT's 0.76 and 0.69), indicating fewer false positives, false negatives, and iden-

Table 3 Comparisons of tracking and counting results based on ground truth (GT) counts

Video sequence ID	Counting method	GT count	Model count	MAE with GT	MOTA
Video 1	SORT	105	459	3.37	0.09
	Deep SORT		255	1.43	0.09
	FairMOT		114	0.09	0.76
	Proposed method		74	0.3	0.79
Video 2	SORT	62	30	0.52	0.12
	Deep SORT		26	0.58	0.14
	FairMOT		42	0.32	0.64
	Proposed method		50	0.19	0.82
Video 3	SORT	52	10	0.81	0.15
	Deep SORT		5	0.9	0.17
	FairMOT		28	0.46	0.69
	Proposed method		25	0.52	0.8
Video 4	SORT	26	16	0.38	0.34
	Deep SORT		11	0.58	0.26
	FairMOT		14	0.46	0.69
	Proposed method		19	0.27	0.84
Video 5	SORT	36	12	0.67	0.27
	Deep SORT		2	0.94	0.25
	FairMOT		27	0.25	0.66
	Proposed method		30	0.17	0.8

MAE Mean absolute error, MOTA Multi-object tracking accuracy

Table 4 Impact of intersection over union (IoU) threshold on detection metrics using enhanced MangoYOLO5

IoU threshold	Precision (%)	Recall (%)	mAP (%)
0.5	89.3	77	91.6
0.6	90.2	73.8	91.3
0.7	91.1	70.4	91.1

mAP Mean average precision

tity switches. This highlights the robustness and reliability of the proposed tracking method, especially in challenging conditions.

Ablation Study

Impact of IoU Threshold on Detection Metrics

To select the best IoU threshold, we conducted an experiment and analyzed its impact on detection results through metrics like precision, recall, and mAP. As shown in Table 4, lowering the IoU threshold from 0.7 to 0.5 significantly improved recall (77.0%) by reducing false negatives. While this adjustment slightly reduced precision (89.3%) due to an increase in false positives. Despite this trade-off, the highest mAP (91.6%) was observed at the 0.5 threshold, demonstrating a robust balance between precision and recall. This finding suggests that a lower IoU threshold is optimal for detecting partially occluded

objects or scenarios where higher recall is essential for performance.

Impact of Architectural Components on Enhanced MangoYOLO5

In this section, we first studied the impact of various attention modules on the proposed detection model, including SE-Net (Hu et al. 2019), ECA-Net (Wang et al. 2020), SK-Net (Li et al. 2019), Granada Attention (Zhao et al. 2024), Mutual-Attention (Transformer) (Zhang and Tian 2023), AFF, and AFF+CBAM, as shown in Table 5. The improved performance of AFF+CBAM is attributed to the effective integration of AFF with CBAM, which enhances feature refinement through channel and spatial attention. The AFF block, comprising CA, SA, C2F, and MSFA, addresses key challenges in mango detection, such as occlusion and scale variation. Specifically, CA enhances channel-specific features to detect mangoes hidden by leaves, SA improves spatial feature representation for irregular clusters, C2F propagates essential features across layers for detecting small mangoes, and MSFA aggregates multi-scale features to handle varying mango sizes and scales. Despite incorporating CBAM, the computational cost of AFF+CBAM remains low at 9.7 giga floating-point operations per second (GFLOPs), making it comparable to other mechanisms, highlighting its efficiency and suitability for real-time applications.

Table 5 Effect of different attention mechanisms in MangoYOLO5

Attention mechanism	Precision (%)	Recall (%)	mAP (%)	FLOPs (G)
SE-Net (Hu et al. 2019)	83.2	70.5	81.2	10.3
ECA-Net (Wang et al. 2020)	83.8	72.6	84.4	10.8
SK-Net (Li et al. 2019)	84.1	73.2	85.1	11.5
Granada Attention (Zhao et al. 2024)	84.7	74	84.9	9.5
Mutual-Attention (Transformer) (Zhang and Tian 2023)	85.2	74.2	86	10.1
AFF	84.3	73.6	85.4	8.2
AFF+ CBAM	85	74.5	86.5	9.7

mAP Mean average precision, *FLOPs* Floating-point operations per second, *AFF* Attentive Feature Fusion, *CBAM* Convolutional Block Attention Module

Table 6 Ablation study results for architectural enhancements in MangoYOLO5

Component	Precision (%)	Recall (%)	mAP (%)
Baseline (MangoYOLO5)	82.8	72.4	83.9
MangoYOLO5 + AFF	84.3	73.6	85.4
MangoYOLO5 + AFF + CBAM	85	74.5	86.5
MangoYOLO5 + AFF + CBAM + Dense Connections	85.5	75.9	87.2
MangoYOLO5 + AFF + CBAM + Dense connections + 10×10	87.8	78.2	89.5
Enhanced MangoYOLO5 model	89.3	77	91.6

mAP Mean average precision

We conducted an ablation study to evaluate the impact of each architectural enhancement in MangoYOLO5, with the results presented in Table 6. The addition of AFF improved feature localization, increasing the mAP by 1.5% over the baseline, particularly addressing challenges such as occlusions and partial visibility of mangoes. Integrating CBAM modules further refined spatial and channel attention, resulting in an additional 1.1% improvement in mAP. Finally, the 10×10 detection layer significantly reduced false negatives, contributing a 2.3% improvement in mAP by handling at varying scales more effectively. The proposed enhanced MangoYOLO5 model combines these enhancements, achieving a total mAP improvement of 7.7% over the baseline, demonstrating its robustness in addressing occlusions, overlapping objects, and scale variations.

Table 7 Ablation analysis of methodological enhancements of the proposed model

Method	Precision (%)	Recall (%)	mAP (%)	MOTA (%)
Baseline (centroid + static thresholding)	89.3	77	91.6	79
IoU for bounding box refinement	90.1	77.3	87.4	80.5
Multi-frame reference tracking	91.2	78.5	91.1	82.1
IoU for bounding box + dynamic threshold tracking	92.5	80.3	92	83.4
Proposed model	94	82	93.6	85.2

mAP Mean average precision, *MOTA* Multi-object tracking accuracy

Impact of Additional Sub-module Components on the Proposed Centroid Tracking Approach

We conducted an extensive ablation analysis to evaluate the impact of different submodules in the proposed tracking approach, using centroid tracking with static thresholding as the baseline. Table 7 highlights the performance improvements achieved with each integrated submodule. For instance, IoU-based bounding box refinement enhances precision and MOTA but slightly reduces mAP due to tighter bounding boxes missing some detections. Multi-frame reference tracking improves recall and robustness by leveraging temporal information but increases computational complexity. The combination of IoU refinement and dynamic threshold tracking boosts all metrics by adapting to varying object sizes, although it requires fine-tuning threshold parameters for optimal results. The proposed model integrates these submodules, achieving the highest performance (mAP of 93.6%), balancing precision, recall, and MOTA. This analysis provides a clear understanding of the trade-offs associated with each submodule.

Conclusions and Future Work

In this work, we present a system for detecting and tracking mangoes to estimate the yield from video sequences captured using stand-alone cameras. The proposed framework consists of an enhanced MangoYOLO5 model, which incorporates attentive feature fusion in the backbone, CBAM and dense connections in the neck, and an additional 10×

10 detect layer to reduce false positives and ensure accurate detection. The tracking module employs a modified centroid tracking algorithm that uses multi-frame reference for occlusion handling, and dynamic threshold selection to uniquely track mangoes across frames. The proposed model achieved an mAP of 93.6% when tested across multiple scenarios involving occlusions and varying lighting conditions, demonstrating its effectiveness in the accurate detection and tracking of mangoes. However, the model's performance in terms of MOTA is observed to be inferior, ranging from a minimum of 76% to a maximum of 84%, when affected by poor video quality factors such as low resolution, blurriness, inconsistent lighting, as well as challenges like extreme occlusion or abrupt camera movements. Future work will focus on incorporating advanced tracking mechanisms and exploring varying imaging distances to address the current limitations of the proposed model, enhancing its robustness and optimization for real-time, large-scale monitoring.

Acknowledgements This research work is funded by TiHAN-IITH under the NM-ICPS scheme, DST Govt. of India, Grant No. TiHAN-IITH/03/2021 – 22/42(4).

Author Contribution Hari Chandana Pichhika: Investigation, Methodology, Software, Validation, Writing—original draft, Data Curation. Priyambada Subudhi: Conceptualization, Supervision, Methodology, Writing—original draft, Visualization. Raja Vara Prasad Yerra: Funding acquisition, Resources, Conceptualization, Methodology, Supervision, Writing—review & editing.

Data availability statement The data that support the findings will be made available upon request after acceptance.

Conflict of interest H.C. Pichhika, P. Subudhi and R.V. Prasad Yerra declare that they have no competing interests.

References

- Anderson N, Underwood J, Rahman M, Robson A, Walsh K (2019) Estimation of fruit load in mango orchards: tree sampling considerations and use of machine vision and satellite imagery. *Precis Agric*. <https://doi.org/10.1007/s11119-018-9614-1>
- Bewley A, Ge Z, Ott L, Ramos F, Upcroft B (2016) Simple online and realtime tracking. *Proc. IEEE Int. Conf. Image Process. (ICIP)*, pp 3464–3468
- Cengel TA, Gencturk B, Yasin ET, Yildiz MB, Cinar I, Koklu M (2024) Apple (*malus domestica*) quality evaluation based on analysis of features using machine learning techniques. *Appl Fruit Sci*: 1–11
- Chaivivatrakul S, Moonrinta J, Dailey MN (2010) Towards automated crop yield estimation—detection and 3d reconstruction of pineapples in video sequences. *Proc. Int. Conf. Comput. Vis. Theory Appl. (VISAPP)*, pp 180–183
- Dai Y, Gieseke F, Oehmcke S, Wu Y, Barnard K (2021) Attentional feature fusion. *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis*, pp 3560–3569
- Delhi PIB (2018) Production and export of mangoes
- Ergün E (2024) Deep learning based multiclass classification for citrus anomaly detection in agriculture. *Signal Image Video Process* 18(11):8077–8088
- Hari Chandana P, Subudhi P, Vara Prasad Yerra R (2023) Mangoyolo5: a fast and compact yolov5 model for mango detection. *Comput. vis. mach. intell.: Proc. cvmi*, vol 2022. Springer, pp 719–731
- Hu J, Shen L, Albanie S, Sun G, Wu E (2019) Squeeze-and-excitation networks (<https://arxiv.org/abs/1709.01507>)
- Junos MH, Mohd Khairuddin AS, Thannirmalai S, Dahari M (2022) Automatic detection of oil palm fruits from uav images using an improved yolo model. *Vis Comput* 38(7):2341–2355
- Kestur R, Meduri A, Narasipura O (2019) Manganet: a deep semantic segmentation architecture for a method to detect and count mangoes in an open orchard. *Eng Appl Artif Intell* 77:59–69
- Li C, Ma W, Liu F, Fang B, Lu H, Sun Y (2023) Recognition of citrus fruit and planning the robotic picking sequence in orchards. *Signal Image Video Process* 17(8):4425–4434
- Li H, Yang M, Yang C, Kang J, Suo X, Meng W, Li Z, Mao L, Sheng B, Qi J (2024) Soccer match broadcast video analysis method based on detection and tracking. *Comput Animat Virtual Worlds* 35(3):e2259
- Li M, Zhang J, Liu H, Yuan Y, Li J, Zhao L (2024) A lightweight method for apple-on-tree detection based on improved yolov5. *Signal Image Video Process*: 1–15
- Li X, Wang W, Hu X, Yang J (2019) Selective kernel networks (<https://arxiv.org/abs/1903.06586>)
- Lin X, Sun S, Huang W, Sheng B, Li P, Feng DD (2021) Eapt: efficient attention pyramid transformer for image processing. *IEEE Trans Multimedia* 25:50–61
- Moonrinta J, Chaivivatrakul S, Dailey MN, Ekpanyapong M (2010) Fruit detection, tracking, and 3d reconstruction for crop mapping and yield estimation. *Proc. IEEE Int. Conf. Control Autom. Robot. Vision*, pp 1181–1186
- Neethi M, Kiran AG, Tiwari H (2021) Yield estimation in mango orchards using machine vision. *AIP Conf Proc* 2336(1):50004
- Neethi M, Kiran AG, Tiwari H (2023) Intelligent mango canopies yield estimation using machine vision. *SN Comput Sci* 4(2):171
- Parico AIB, Ahamed T (2021) Real-time pear fruit detection and counting using yolov4 models and deep sort. *Sensors* 21(14):4803
- Payne AB, Walsh KB, Subedi P, Jarvis D (2013) Estimation of mango crop yield using image analysis–segmentation method. *Comput Electron Agric* 91:57–64
- Prakash AJ, Prakasam P (2023) An intelligent fruits classification in precision agriculture using bilinear pooling convolutional neural networks. *Vis Comput* 39(5):1765–1781
- Ren S, He K, Girshick R, Sun J (2015) Faster r-cnn: towards real-time object detection with region proposal networks. *Adv Neural Inf Process Syst* 28:
- Santos TT, de Souza LL, dos Santos AA, Avila S (2020) Grape detection, segmentation, and tracking using deep neural networks and three-dimensional association. *Comput Electron Agric* 170:105247
- Sharma A, Kumar V, Longchamps L (2024) Comparative performance of yolov8, yolov9, yolov10, yolov11 and faster r-cnn models for detection of multiple weed species. *Smart Agric Technol* 9:100648
- Si H, Wang Y, Liu Q, Li W, Wan L, Song J, Zhao W, Sun C (2024) A registration algorithm for the infrared and visible images of apple based on active contour model. *Vis Comput* 40(4):2833–2855
- Stein M, Bargouti S, Underwood J (2016) Image-based mango fruit detection, localisation and yield estimation using multiple view geometry. *Sensors* 16(11):1915
- Ultralytics (2021) Yolov5 (<https://github.com/ultralytics/yolov5>)
- Wang L, Zhao Y, Xiong Z, Wang S, Li Y, Lan Y (2022) Fast and precise detection of litchi fruits for yield estimation based on the improved yolov5 model. *Front Plant Sci* 13:965425
- Wang Q, Wu B, Zhu P, Li P, Zuo W, Hu Q (2020) Ecanet: efficient channel attention for deep convolutional neural networks. *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*

- Wang Z, Walsh K, Koirala A (2019) Mango fruit load estimation using a video based mangoyolo—kalman filter—Hungarian algorithm method. *Sensors* 19(12):2742
- Wojke N, Bewley A, Paulus D (2017) Simple online and realtime tracking with a deep association metric. *Proc. IEEE Int. Conf. Image Process. (ICIP)*, pp 3645–3649
- Woo S, Park J, Lee J-Y, Kweon IS (2018) Cbam: convolutional block attention module. *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pp 3–19
- Zhang M, Tian X (2023) Transformer architecture based on mutual attention for image-anomaly detection. *Virtual Real Intell Hardw* 5(1):57–67
- Zhang W, Wang J, Liu Y, Chen K, Li H, Duan Y, Wu W, Shi Y, Guo W (2022) Deep-learning-based in-field citrus fruit detection and tracking. *Hortic Res* 9:
- Zhang Y, Wang C, Wang X, Zeng W, Liu W (2021) Fairmot: on the fairness of detection and re-identification in multiple object tracking. *Int J Comput Vis* 129:3069–3087
- Zhao J, Du C, Li Y, Mudsh M, Guo D, Fan Y, Wu X, Wang X, Almodfer R (2024) Yolo-granada: a lightweight attentioned yolo for pomegranates fruit detection. *Sci Rep* 14(1):16848

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.